Candidate name:

_____

Date: _____

Unofficial NUFYPET Math Simulation

Time: 1 hour (60 min.) Version 11.1
Total Marks: ____/30

Instructions

- Circle the answer you find right
- Calculators **MUST NOT** be used
- Figures are **NOT** accurately drawn and **NOT** to scale, except it is mentioned in a question
- There is only one answer for each question, so answers with two or more selected options marked as zero
- If you think something is wrong with the test, report a Methodist

Information

- The total mark for test is 30 (Each question marked as 1 point)
- There is **NO** penalty for wrong answers

Advice

- Read each question carefully
- Follow the time (There is an average 2 minutes for each question)
- Don't stay in one task for long
- Try to answer for all questions
- Recheck your answers if possible

Note: This is unofficial test made by MSDC

1. Which of the following is NOT equal to 0.0675?

A) 67.5×10^{-3}

B) 6.75×10^{-2}

C) 0.675×10^{-1}

D) 0.00675×10^{2}

E) 0.0000675×10^{3}

2. A polynomial $p(x)$ has the property that $p(1) = 2$.

Which one of the following can be deduced from this?

A) $p(x) = (x-1)q(x) + 2$ for some polynomial $q(x)$.

B) $p(x) = (x+1)q(x) + 2$ for some polynomial $q(x)$.

C) $p(x) = (x-1)q(x) - 2$ for some polynomial $q(x)$.

D) $p(x) = (x-1)q(x) - 2$ for some polynomial $q(x)$.

E) $p(x) = (x-2)q(x) + 1$ for some polynomial $q(x)$.

F) $p(x) = (x+2)q(x) + 1$ for some polynomial $q(x)$.

G) $p(x) = (x-2)q(x) - 1$ for some polynomial $q(x)$.

H) $p(x) = (x+2)q(x) - 1$ for some polynomial $q(x)$.

3. A quadratic equation has a solutions a and b

$$a + b = -5$$

$$ab = 3$$

What is the equation?

A) $x^2 + 5x + 3 = 0$ B) $x^2 - 5x - 3 = 0$ C) $x^2 - 5x + 3 = 0$
D) $x^2 + 3x + 5 = 0$ E) $x^2 + 5x - 3 = 0$ F) $x^2 + 3x - 5 = 0$

4. The total surface area of a cylinder is numerically the same as its volume. The radius of the cylinder is r cm, the height is h cm.

Express h in terms of r.

A) $h = \frac{2r}{r-2}$ B) $h = \frac{2r}{r+2}$ C) $h = r + 2$ D) $h = r - 2$ E) $h = 2r(r-2)$

5. A bag contains 8 blue and 2 red balls. Three balls are chosen at the same time at random from the bag. What is the probability that exactly two of the balls are the same colour?

A) $\frac{5}{8}$ B) $\frac{4}{9}$ C) $\frac{8}{15}$ D) $\frac{3}{8}$ E) $\frac{7}{15}$ F) $\frac{6}{7}$

6. In the formula $P = 2r^2t$

$$r = 3 \times 10^{-3}$$

$$t = 2.5 \times 10^{4}$$

Calculate the value of P. Leave your answer in standard form.

A) 7.5×10^{-2} B) 45×10^{-2} C) 75×10^{-2} D) 4.5×10^{-1} E) 7.5×10^{-1} F) 4.5

7. Askar has a collection of 5-tenge and 10-tenge coins worth a total of 400 tenge. If he has 50 coins in all, how many 5-tenge coins does Askar have?

A) 15 B) 20 C) 25 D) 30 E) 40

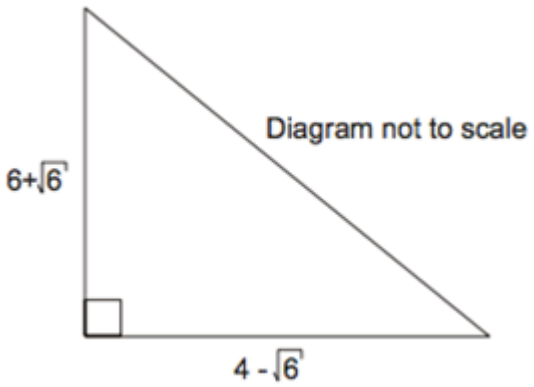
8. The longest side of a right angled triangle is $6 + \sqrt{5}$ units. One of the shorter sides is $3 + 2\sqrt{5}$ units.

What is the length of the third side?

A) $2\sqrt{3}$ B) $\sqrt{70 + 24\sqrt{5}}$ C) 12 D) $3 - \sqrt{5}$ E) $14 + 7.5\sqrt{5}$

9. A right-angled triangle is shown.

Calculate the area of the triangle.



A) $9 - \sqrt{6}$ B) $9 - \sqrt{3}$ C) $\sqrt{52 + 4\sqrt{6}}$ D) $9 + \sqrt{6}$ E) $15 + 5\sqrt{6}$

10. The sequence a_n is given by the rule:

$$a_1 = 2$$

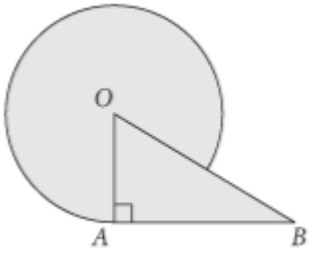
$a_{n+1} = a_n + (-1)^n$ for $n \geq 1$

What is

$$\sum_{n=1}^{100} a_n$$

A) 150 B) 250 C) -4750
D) 5150 E) $4(1 - \left(\frac{1}{2}\right)^{100})$ F) $4(\left(\frac{3}{2}\right)^{100} - 1)$

11. In the figure, if radius OA is 8 and the area of right triangle OAB is 32, what is the area of the shaded region?



A) $64\pi + 32$ B) $60\pi + 32$ C) $56\pi + 32$ D) $32\pi + 32$ E) $16\pi + 32$

12. What will be the answer if the smallest two-digit odd number is subtracted from the greatest three-digit even number?

A) -987 B) -989 C) 989 D)) 988 E) 987

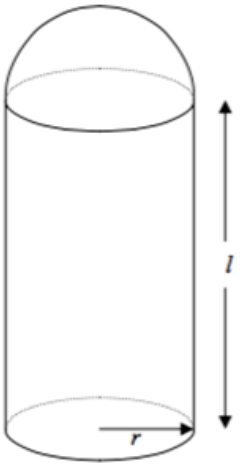
13. In 2001 a quarter of the population of Wales reported having a limiting long-term illness or disability which restricted their daily activities.

What is the probability that out of a random group of three people who live in Wales, exactly one of them will have reported such an illness or disability?

A) $\frac{9}{64}$ B) $\frac{27}{64}$ C) $\frac{37}{64}$ D) $\frac{3}{4}$

14. The solid shown below consists of a cylinder topped by a hemisphere of the same radius.

Which one of the following correctly gives the volume of this solid?



A) $\frac{\pi r^2}{3}(2r+3l)$ B) $\frac{\pi r^2}{3}(4r+3l)$ C) $\frac{\pi r^2}{3}(2+3l)$ D) $\frac{\pi r^2}{3}(2r+l)$

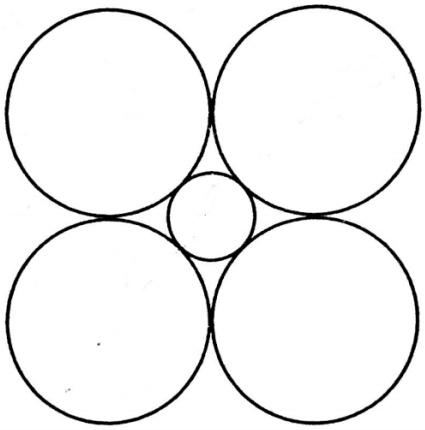
15. If $3^{n-2}+3^2=36$, find n.

A) 3 B) 4 C) 5 D) 6 E) 7

16. Four consecutive are such that three times the smallest is two times the largest. What is the largest of these four integers?

A) 7 B) 9 C) 6 D) 11 E) 4

17. The total area of the four equal circles in the figure above is 36π, and the circles are all tangent to one another. What is the diameter of the small circle?



A) $6\sqrt{2}$ B) $6+\sqrt{2}$ C) $3\sqrt{2}-3$ D) $6\sqrt{2}-6$ E) $6\sqrt{2}+6$

18. Three variables x, y and z are known to be related to each other in the following ways:

- X is directly proportional to the square of z
- Y is inversely proportional to the cube of z.

Which of the following correctly describes the relationship between x and y?

A) The square of x is directly proportional to the cube of y.

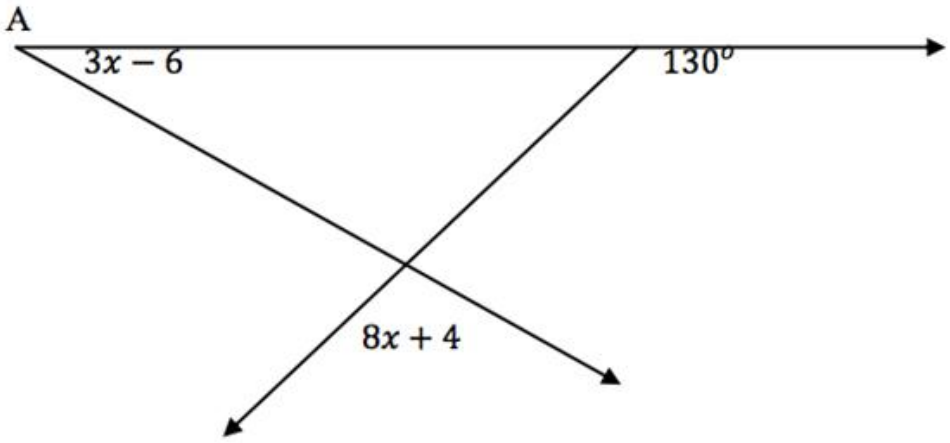
B) The square of x is inversely proportional to the cube of y.

C) The cube of x is directly proportional to the square of y.

D) The cube of x is inversely proportional to the square of y.

E) x is directly proportional to y^6.

19. Find angle at A.



A) 30 B) 12 C) 24 D) 50 E) 100

20. If A percent of B is C, the C is what percent of AB?

A) 100% B) AB/C% C) C/AB% D) 1% E) Undefined

21. A rectangle with integer side length has perimeter 10. What is the greatest number of these rectangles that can be cut from a piece of paper with width 24 and length 60?

A) 144 B) 180 C) 240 D) 360 E) 480

22. If $q^p = 4 - qp$ and $p^q = 1$, where q and p are positive integers, what is the value of q

A) 0 B) 1 C) 2 D) 3 E) 4

23. The total cost of 20 identical books is x dollars. At this rate, what is the total cost in dollars of 70 of these books in terms of x?

A) $7x/2$ B) $5x/7$ C) $2x/7$ D) $7x/5$ E) $70x$

24. I have two six-sided dice, each with faces numbered from 1 to 6. One of the dice is fair, but the other is not – it will land on numbers 1 to 5 with equal probability, but lands on 6 with a different probability.

When I roll the dice the probability that I get a total of 12 is $\frac{1}{18}$.

What is the probability that I get a total of 2 when I roll the dice?

A) $\frac{1}{72}$ B) $\frac{1}{45}$ C) $\frac{1}{36}$ D) $\frac{1}{18}$ E) $\frac{1}{9}$

25. Triangles ABC and XYZ have the same area.

Which of these extra conditions, taken independently, would imply that they are congruent?

(1) $AB = XY$ **and** $BC = YZ$

(2) $AB = XY$ **and** $\angle ABC = \angle XYZ$

(3) $\angle ABC = \angle XYZ$ **and** $\angle BCA = \angle YZX$

	Condition (1)	Condition (2)	Condition (3)
A	Does not imply congruent	Does not imply congruent	Does not imply congruent
B	Does not imply congruent	Does not imply congruent	Implies congruent
C	Does not imply congruent	Implies congruent	Does not imply congruent
D	Does not imply congruent	Implies congruent	Implies congruent
E	Implies congruent	Does not imply congruent	Does not imply congruent
F	Implies congruent	Does not imply congruent	Implies congruent
G	Implies congruent	Implies congruent	Does not imply congruent
H	Implies congruent	Implies congruent	Implies congruent

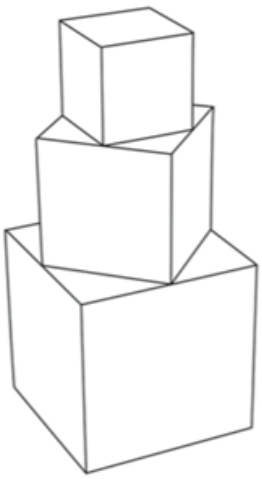
26. At a fancy vegetable stand, Tom bought purple tomatoes for \$8 each and Jerry bought organic giant cabbages \$12 each. In total, they paid \$104 for 10 vegetables. How many cabbages did Jerry buy?

A) 4 B) 5 C) 6 D) 7 E) 3

27. A right circular cone and a sphere have equal volumes. If the cone has radius x and height $2x$, what is the radius of the sphere?

A) x B) $\frac{x}{\sqrt[3]{2}}$ C) $\sqrt[3]{2}$ D) $\frac{1}{\sqrt[3]{2}}$ E) $\sqrt[3]{2}x$

28. A solid shape is made by joining three cubes together with the largest cube on the bottom and the smallest on the top. Where the faces of two cubes join, the comers of the smaller cube are at the midpoints of the sides of the larger cube.



The sides of the smallest cube have a length of $1\ cm$. What is the total surface area of the shape?

A) $30\ cm^2$ B) $32\ cm^2$ C) $33\ cm^2$
D) $36\ cm^2$ E) $39\ cm^2$ F) $42\ cm^2$

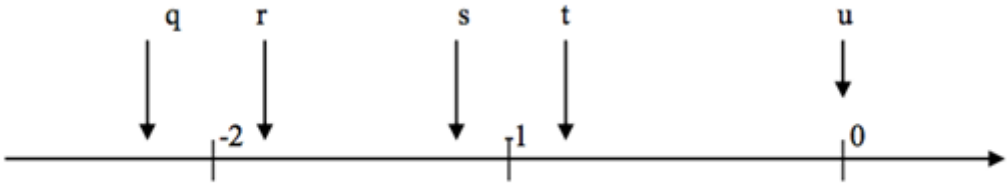
29. A group of five numbers are such that:

- Their mean is 0
- Their range is 20

What is the largest possible median of the five numbers?

A) 0 B) 4 C) $4\frac{1}{2}$ D) $6\frac{1}{2}$ E) 8 F) 20

30. The numbers q, r, s, t and u are indicated in the number line above. Which of the following is the largest value?



A) qt B) qu C) rs D) qr E) st

Answer key:

№	Correct Answer	Your answer
1	D	
2	A	
3	A	
4	A	
5	C	
6	D	
7	B	
8	A	
9	A	
10	C	
11	C	
12	E	
13	B	
14	A	
15	C	
16	B	
17	D	
18	B	
19	A	
20	D	
21	C	
22	C	
23	A	
24	B	
25	H	
26	C	
27	B	
28	D	
29	E	
30	D	
		Total: _____